

RUBEN DARIO FLOREZ ZELA

Electronic Engineer · RENACYT Level V Researcher

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ORCID | Google Scholar | ResearchGate | LinkedIn | Personal Web

Citations: 143 · h-index: 5 · i10-index: 3 (Google Scholar, 2025)

RESEARCH PROFILE

Human-centered AI researcher specializing in multimodal systems for real-time driver-state monitoring in intelligent transportation environments. My work integrates computer vision (RGB–NIR imaging), gaze analysis, and deep learning to detect driver drowsiness, distraction, and cognitive workload under real-world conditions. I design and deploy low-latency AI systems on embedded platforms (NVIDIA Jetson Nano) for on-vehicle inference in safety-critical scenarios.

I am recognized as a RENACYT Level V researcher by CONCYTEC (Peru's National Science Council) and serve as a peer reviewer for international journals including *Scientific Reports* (Nature Portfolio) and *ACM Transactions on Intelligent Systems and Technology*. My research addresses a high-impact public safety challenge: driver-related cognitive impairment is a leading factor in road traffic accidents worldwide.

My work bridges multimodal perception and human-centered system design for safer automated driving. I am seeking a PhD position to advance human-centered AI, adaptive intelligent systems, and transportation safety.

EDUCATION

M.Sc. in Electronic Engineering (Automation & Instrumentation)

2024 – Present

Universidad Nacional de San Agustín de Arequipa (UNSA), Arequipa, Peru

Thesis focus: A multimodal framework for driver-state monitoring in automated driving using computer vision and embedded AI systems

B.Sc. in Electronic Engineering

Graduated

Universidad Nacional de San Antonio Abad del Cusco (UNSAAC), Cusco, Peru

Professional License (CIP): 336757

SELECTED PEER-REVIEWED JOURNAL ARTICLES

- [1] **Florez, R.**, Palomino-Quispe, F., Alvarez, A. B., Coaquira-Castillo, R. J., & Herrera-Levano, J. C. (2024). A Real-Time Embedded System for Driver Drowsiness Detection Based on Visual Analysis of the Eyes and Mouth Using Convolutional Neural Network and Mouth Aspect Ratio. *Sensors*, 24(19), 6261. [27 citations]
- [2] **Florez, R.**, Palomino-Quispe, F., Coaquira-Castillo, R. J., Herrera-Levano, J. C., Paixão, T., & Alvarez, A. B. (2023). A CNN-Based Approach for Driver Drowsiness Detection by Real-Time Eye State Identification. *Applied Sciences*, 13(13), 7849. [82 citations — #1 most cited]
- [3] Fuentes-Beingolea, J. H., Palomino-Quispe, F., Herrera-Levano, J. C., Vargas-Mateos, W., **Florez, R.**, & Alvarez, A. B. (2025). Illumination-Robust Conjunctival Image Preprocessing for Accurate Segmentation and Anemia Detection Using Deep Learning. *International Journal of Online and Biomedical Engineering (iJOE)*, 21(07), 106–124.
- [4] Fragoso, J., Silva, C., Paixão, T., Alvarez, A. B., Júnior, O. C., **Florez, R.**, Palomino-Quispe, F., Savian, L. G., & Trazzi, P. A. (2025). Coffee-Leaf Diseases and Pests Detection Based on YOLO Models. *Applied Sciences*, 15(9), 5040. [14 citations]

- [5] Jancco-Chara, J., Palomino-Quispe, F., Coaquira-Castillo, R. J., Herrera-Levano, J. C., & **Florez, R.** (2023). Doppler Factor in the Omega-k Algorithm for Pulsed and Continuous Wave Synthetic Aperture Radar Raw Data Processing. *Applied Sciences*, 14(1), 320. [6 citations]
- [6] Paixão, T., Alvarez, A. B., **Florez, R.**, & Palomino-Quispe, F. (2023). Fuzzy Controller Implemented for Movement of a Tendon-Driven 3D Robotic Lumbar Spine Mechanism. *Sensors*, 23(24), 9633. [4 citations]

SELECTED PEER-REVIEWED CONFERENCE PAPERS

- [1] Paixão, T., Alvarez, A. B., **Florez, R.**, & Palomino-Quispe, F. (2023). Fuzzy Control for Simplified Lumbar Spine Robotic Mechanism Motion. *2023 IEEE International Conference on Networking, Sensing and Control (ICNSC)*, Vol. 1, pp. 1–6. IEEE.
- [2] Paixão, T., Alvarez, A. B., **Florez, R.**, & Palomino-Quispe, F. (2023). Motion Control of a Robotic Lumbar Spine Model. *Int'l Work-Conference on Bioinformatics and Biomedical Engineering (IWBBIO)*, pp. 205–216. Springer Nature.
- [3] Paixão, T., Alvarez, A. B., **Florez, R.**, Palomino-Quispe, F., Angst, L., & Maggi, L. (2023). Development of Simplified Lumbar Spine Mechanism Implemented with Tendon-Driven Motion. *2023 27th International Conference on Methods and Models in Automation and Robotics (MMAR)*, pp. 187–192. IEEE.
- [4] **Florez, R.**, Concha-Ramos, Y., Palomino-Quispe, F., & Coaquira-Castillo, R. J. (2022). CNN for the Detection of COVID-19 from Chest X-Ray Images. *2022 IEEE Engineering International Research Conference (EIRCON)*, pp. 1–4. IEEE.
- [5] Vilavila, R., Justo, F., **Florez, R.**, & Figueroa, N. (2021). Low-Cost Mini Humanoid Robot Mechanical Design for Mini Humanoid Robot Contest INTERCON 2021. *2021 IEEE XXVIII International Conference on Electronics, Electrical Engineering and Computing (INTERCON)*, pp. 1–4. IEEE.

RESEARCH PROJECTS

Driver-State Monitoring via Embedded Vision (Principal Investigator)
LIECAR Lab (UNSAAC) & PAVIC Lab (Federal University of Acre – UFAC, Brazil)

Feb 2023 – Aug 2023

- Designed a novel ROI correction algorithm for eye-region detection under varying illumination conditions
- Developed and benchmarked 3 CNN architectures via transfer learning for eye-state classification
- Proposed a custom CNN architecture for real-time driver monitoring; deployed on NVIDIA Jetson Nano

Control of a Simplified Lumbar Spine Mechanism (Co-Investigator)
LIECAR Lab (UNSAAC) & PAVIC Lab (UFAC, Brazil)

Feb 2023 – Jul 2023

- Developed PID and Fuzzy Logic controllers for vertebral movement in a tendon-driven 3D-printed lumbar spine model

Computer Vision Applied to a Low-Cost Mini Humanoid Robot (Collaborator)
RoboticsLab (UNSAAC) & CONCYTEC

Mar 2021 – Apr 2022

- Implemented a computer vision guidance system enabling autonomous navigation in competition environments

ACADEMIC POSITIONS

Contract Lecturer – Electronic Engineering Department
Universidad Nacional de San Antonio Abad del Cusco (UNSAAC), Cusco, Peru

April 2024 – Present

- Courses: Robotics, Digital Image Processing, Artificial Intelligence, Electronics Laboratory
- Teaching undergraduate students in embedded systems, artificial intelligence, and computer vision with application-oriented laboratory activities

PEER REVIEW SERVICE

Active reviewer for the following indexed journals:

- ACM Transactions on Intelligent Systems and Technology
- Scientific Reports (Nature Portfolio)
- Discover Artificial Intelligence (Springer Nature)
- Discover Applied Sciences (Springer Nature)
- Discover Internet of Things (Springer Nature)
- International Journal of Science and Technology (IJST)
- IAES International Journal of Artificial Intelligence (IJ-AI)

RESEARCH INTERESTS

Driver State Monitoring · Human-Centered AI · Computer Vision (RGB–NIR) · Multimodal AI Systems · Gaze Analysis and Eye-Tracking · Deep Learning for Real-Time Systems · Embedded AI (NVIDIA Jetson) · Intelligent Transportation Systems · Transportation Safety · Automated Driving

AWARDS & RECOGNITIONS

1st Place – VIII Regional Science, Technology & Innovation Fair Sep 2023
Regional Government of Cusco, OCTI & CORCYTEC | Project: COVID-19 Diagnostic System via AI

1st Place – II International Congress of Research, Innovation & Entrepreneurship Nov 2022
UNSAAC | Project: Real-Time Driver Drowsiness Detection System

1st Place – First Andean Region Hackathon Feb 2023
UNI, Dara Big Data & Hack4Dev (Venezuela, Peru, Colombia, Ecuador, Bolivia, Chile) | Galaxy Classification via ML

1st Place – Mini Humanoid Robot Contest (×2) 2021
CONEIMERA UNA PUNO 2021 | INTERCON 2021

2nd Place – Hult Prize (On-Campus Round) Jan 2020
UNSAAC | Advanced to Regional Finals in Bogotá, Colombia

INVITED TALKS & PRESENTATIONS

Drowsiness Detection System for Vehicle Drivers Using Neural Networks May 2023
VII Seminar on Industrial Electronics and Automotive Mechatronics | IESTP EPAMET

Python in Image Processing and Artificial Intelligence (Webinar) Jun 2024
Pythonistas Community

Machine Vision and AI in Robotics (Webinar) Oct 2021

Computer Vision and AI Applied to Humanoid Robots (Webinar)
IEEE Computer Society – UDEP Lima & Data Science Research Peru

Oct 2021

Deep Learning for COVID-19 Diagnosis and Other Applications (Webinar)
IEEE RAS – UNSAAC

Apr 2021

PROFESSIONAL MEMBERSHIPS

- IEEE Engineering in Medicine and Biology Society (IEEE EMBS)
- RENACYT Researcher Level V – CONCYTEC, Peru

TECHNICAL SKILLS

Programming	Python, MATLAB, C/C++ (Arduino / STM)
Deep Learning & ML	TensorFlow, Keras, OpenCV, Pandas, NumPy, Matplotlib
Computer Vision	RGB–NIR camera systems, ROI extraction, gaze analysis, facial landmark tracking
Embedded Platforms	NVIDIA Jetson Nano, Raspberry Pi, ESP-32, STM32, Arduino
Hardware	PCB design & fabrication, Oscilloscope, Multimeter
CAD / Simulation	Autodesk Fusion 360, Proteus, Eagle

LANGUAGES

Spanish	Native
English	Intermediate (academic reading/writing; improving spoken)

INTERNATIONAL RESEARCH COLLABORATIONS

- Federal University of Acre (UFAC), Brazil – Joint projects in embedded vision and robotic control systems (PAVIC Lab)